

HTML Links & Images

Complete Reference with Syntax, Attributes & Examples

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This guide is a thorough reference for two of the most-used HTML elements: **<a>** (anchor / hyperlink) and **** (image). Each section explains the tag, lists every important attribute with examples, shows the rendered output, and highlights best practices for accessibility and SEO.

1. Introduction to Links and Images

Two HTML elements are responsible for almost every bit of navigation and visual content on the web:

Anchor / Link	<code><a></code>	Navigate to another page, file, email, or section	Inline
Image	<code></code>	Embed a raster or vector image into the page	Inline (void)

Void element: `<img>` has no closing tag — it is self-closing. In HTML5 you write `<img src='...' alt='...'/>`; in XHTML you write `<img src='...' alt='...' />`.

2. HTML Links — <a> (Anchor Tag)

The **<a>** (anchor) tag creates a hyperlink. It can link to another webpage, a specific section on the same page, a file download, an email address, or a phone number.

2a. Basic Syntax

```
<a href="URL">Link Text</a>
```

The content between the opening and closing **<a>** tags is what the user sees and clicks. The **href** attribute holds the destination.

2b. Anchor Tag Attributes

Attribute	Value / Syntax	Description
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href	URL / path / #id / mailto: / tel:	Specifies the link destination (HyperText REference). Required for a clickable link.
target	_self _blank _parent _top framename	Where to open the linked document. Default is _self (same tab).
rel	noopener noreferrer nofollow alternate ...	Describes the relationship between the current page and the linked page.
title	text string	Tooltip text shown on hover.
download	filename (optional)	Prompts the browser to download the linked file instead of navigating to it.
hreflang	language code e.g. en, fr, te	Specifies the language of the linked document.
type	MIME type e.g. text/html	Hints at the content type of the linked resource.
id / class / style	any valid value	Standard global attributes for styling and scripting.

2c. Link Examples

External Link

```
<a href="https://www.example.com">Visit Example</a>
```

■ Visit Example (opens same tab)

Open in New Tab

```
<a href="https://www.example.com"
  target="_blank"
  rel="noopener noreferrer">Open in New Tab</a>
```

■ Open in New Tab (new tab)

Email Link

```
<a href="mailto:hello@example.com">Send Email</a>
```

✉ hello@example.com (opens mail client)

Phone Link

```
<a href="tel:+919876543210">Call Us</a>
```

■ Call Us (+91 98765 43210)

Download Link

```
<a href="resume.pdf" download="MyResume">
  Download Resume
</a>
```

■ Download Resume (saves resume.pdf)

Anchor (Same Page)

```
<!-- Target section -->
<h2 id="contact">Contact</h2>

<!-- Link to it -->
<a href="#contact">Go to Contact</a>
```

■ Go to Contact (jumps to #contact section)

2d. target Attribute Values Explained

Value	Behaviour
<code>_self</code>	Default. Opens in the same tab/window.
<code>_blank</code>	Opens in a new tab or window.
<code>_parent</code>	Opens in the parent frame (used with iframes).
<code>_top</code>	Opens in the full browser window, breaking out of all frames.
<code>framename</code>	Opens in a named <iframe> or <frame>.

Security note: Always add `rel="noopener noreferrer"` when using `target="_blank"`. Without it, the opened page can access the opener's window object — a potential phishing risk.

2e. Types of URL Paths

Absolute URL	<code>https://www.site.com/page</code>	Linking to an external site
Root-relative	<code>/images/photo.jpg</code>	Same domain; path from root
Relative path	<code>../about/contact.html</code>	Navigate relative to the current file
Fragment (#id)	<code>#section2</code>	Jump to an element on the same page
mailto:	<code>mailto:user@example.com</code>	Open default email client
tel:	<code>tel:+910000000000</code>	Dial a phone number (mobile)

3. HTML Images — (Image Tag)

The **** tag embeds an image into an HTML document. It is a **void (self-closing)** element — it has no content and no closing tag. At minimum, every **** must have a **src** and an **alt** attribute.

3a. Basic Syntax

```

```

3b. Image Tag Attributes

Attribute	Value / Syntax	Description
src	URL / file path	Source of the image. Can be absolute, relative, or a data URI. REQUIRED.
alt	text string	Alternative text for accessibility and SEO. Shown if image fails to load. REQUIRED.
width	pixels or %	Display width of the image. Specify both width & height to prevent layout shift.
height	pixels or %	Display height of the image.
title	text string	Tooltip shown on hover.
loading	lazy eager	lazy defers loading until the image enters the viewport — improves performance.
decoding	async sync auto	Hint for the browser's image decoding strategy.
srcset	url 1x, url 2x ...	Defines multiple image sources for responsive images at different resolutions.
sizes	media condition + width	Works with srcset to specify which source to use at different viewport sizes.
usemap	#mapname	Associates the image with a client-side image map (see Section 5).
ismap	(boolean)	Marks the image as a server-side image map.
crossorigin	anonymous use-credentials	Handles CORS requests for the image.
style / class / id	any valid value	Standard global attributes for styling and scripting.

3c. Image Examples

Basic Image

```

```

[photo.jpg]

A scenic mountain view

Image with Width & Height

```

```

[logo.png — 200×80 px]

Company Logo

Image as a Link

```
<a href="https://example.com">
  
</a>
```

■ [banner.jpg] — clicking opens example.com

Lazy Loading

```

```

Image loads only when scrolled into view (saves bandwidth)

Image from External URL

```

```

[Placeholder — 400×200]

3d. Supported Image Formats

Format	Extension	Best For	Transparency
JPEG	.jpg / .jpeg	Photos, complex images	No
PNG	.png	Logos, icons, graphics with sharp edges	Yes (alpha)
GIF	.gif	Simple animations, limited colours	Yes (1-bit)
SVG	.svg	Icons, logos — scales to any size	Yes
WebP	.webp	Modern format — smaller than JPEG/PNG	Yes

AVIF	.avif	Next-gen — best compression	Yes
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4. Responsive Images — srcset & sizes

Modern devices have varying screen densities (1x, 2x, 3x). The **srcset** and **sizes** attributes let the browser choose the most appropriate image, saving bandwidth on small screens and delivering sharp images on high-DPI displays.

4a. Resolution Switching (srcset with descriptors)

```

```

srcset lists available files with their actual pixel widths (400w, 800w ...). **sizes** tells the browser how wide the image will be displayed at different viewport sizes. The browser picks the most appropriate source automatically.

4b. Art Direction with <picture>

Use <picture> when you need a different *crop* or *composition* at different breakpoints, not just a different resolution.

```
<picture>
  <!-- Wide screen: landscape crop -->
  <source media="(min-width: 800px)"
    srcset="hero-wide.jpg">
  <!-- Tablet: square crop -->
  <source media="(min-width: 480px)"
    srcset="hero-square.jpg">
  <!-- Fallback for all browsers -->
  
</picture>
```

5. Image Maps — <map> & <area>

An **image map** makes different regions of a single image clickable, each linking to a different URL. Three tags work together: ****, **<map>**, and **<area>**.

```


<map name="worldmap">

  <!-- Rectangle area -->
  <area shape="rect"
        coords="10,20,150,100"
        href="europe.html"
        alt="Europe">

  <!-- Circle area -->
  <area shape="circle"
        coords="300,150,60"
        href="asia.html"
        alt="Asia">

  <!-- Polygon area -->
  <area shape="poly"
        coords="400,50,500,80,480,150,390,130"
        href="africa.html"
        alt="Africa">

</map>
```

shape	coords format	Description
rect	x1,y1,x2,y2	Top-left and bottom-right corners of a rectangle
circle	cx,cy,radius	Centre x, centre y, and radius
poly	x1,y1,x2,y2,x3,y3...	List of x,y pairs defining the polygon vertices
default	(none)	Entire image — acts as a fallback if no other area matches

6. Accessibility & SEO Best Practices

6a. Writing Good alt Text

The **alt** attribute is the most important accessibility attribute for images. Screen readers read it aloud for visually impaired users. Search engines use it to understand image content.

Situation	alt Value	Reason
Informative image	<code>alt="Golden retriever puppy sitting on grass"</code>	Describes what the image shows
Decorative image	<code>alt=""</code> (empty string)	Screen reader skips it — image adds no meaning
Linked image (icon)	<code>alt="Home"</code>	Describes the link destination, not the icon shape
Complex chart	<code>alt="Bar chart showing sales..."</code>	Brief description; add long description elsewhere

6b. Links Accessibility Checklist

- Use descriptive link text (not 'click here')
- Add `rel="noopener noreferrer"` for `target="_blank"`
- Ensure sufficient colour contrast for link text
- Use `:focus` styles so keyboard users see where they are

- Avoid opening new tabs unexpectedly
- Use title attribute to add extra context on hover
- Test links with a screen reader
- Group navigation links inside `<nav>`

6c. Images Accessibility Checklist

- Every `<img>` must have an alt attribute
- Decorative images use `alt=""` (empty)
- Specify width and height to prevent CLS
- Use `loading="lazy"` for below-the-fold images

- Use descriptive file names (dog-sitting.jpg)
- Compress images — use WebP/AVIF for modern browsers
- Provide text alternatives for complex images
- Never use images of text (use real HTML text)

7. Complete HTML Page — Links & Images Together

The following example combines everything covered: external links, anchor links, image embedding with alt text, lazy loading, a linked image, and a mailto link.


```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Photo Blog</title>
</head>
<body>

  <!-- Navigation with anchor links -->
  <nav>
    <a href="#gallery">Gallery</a> |
    <a href="#contact">Contact</a> |
    <a href="about.html">About</a>
  </nav>

  <!-- Linked logo image -->
  <a href="index.html">
    
  </a>

  <h1>Welcome to My Photo Blog</h1>
  <p>Explore stunning photography from around the world.</p>

  <!-- Image gallery section -->
  <section id="gallery">
    <h2>Featured Photos</h2>

    <a href="photo1-full.jpg">
      
    </a>

    <a href="photo2-full.jpg">
      
    </a>

  </section>
</body>
</html>
```

```

</section>

<!-- External link -->
<p>
  Learn more at
  <a href="https://photography.example.com"
    target="_blank"
    rel="noopener noreferrer">
    Photography Tips
  </a>.
</p>

<!-- Contact section with mailto -->
<section id="contact">
  <h2>Contact</h2>
  <p>
    <a href="mailto:photo@example.com">photo@example.com</a> |
    <a href="tel:+910000000000">+91 00000 00000</a>
  </p>
</section>

</body>
</html>

```

8. Quick Reference Card

Basic link	<code>Text</code>
New tab link	<code>Text</code>
Email link	<code>Email</code>
Phone link	<code>Call</code>
Download link	<code>Download</code>
Anchor (same page)	<code>Jump + <h2 id="id"></code>
Basic image	<code></code>
Image with size	<code></code>

Lazy load image	<code></code>
Linked image	<code></code>
Responsive image	<code></code>

HTML Links & Images Reference · Practice by building real pages — experiment freely!